A Hybrid Neural Approach to Speaker Verification with an Improved Additive Angular Margin Loss

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***Abstract*—Extraction of speaker embeddings plays a crucial role in the neural automatic speaker verification system. Here, we propose a novel hybrid neural embedding framework, which employs frequency- and channel-wise Selective Kernel Attention (SKA) into the 2D-CNN - based feature extraction module to aggregate global frequency-channel information to the attention weights to extract speaker discriminant embeddings. The aforementioned feature extraction module is connected with a frame-level network, which is composed of a Time Delay Neural Network (TDNN)-Long Short Term Memory hybrid network and a fully TDNN network in a cascade fashion. Multi-Level Attentive Statistics Pooling, which incorporates local statistics as context, is adopted for aggregating the speaker information within an utterance-level context by capturing the complementarity of different networks. Additionally, the proposed approach utilizes an improved Additive Angular Margin (AAM) Softmax loss function that integrates a dynamic and adaptive label noise cleansing method, termed AdaptiveDrop. This method seamlessly combines label noise filtering and correction in a cascaded manner, leveraging the strengths of both techniques to enhance robustness. Experimental results on the VoxCeleb dataset reveal that the proposed approach outperforms baseline systems in both supervised and self-supervised speaker verification tasks.**

***Index Terms*—speaker verification, hybrid network, local statistics, selective kernel attention, multi-level ASP.**

I. Introduction

Speaker verification (SV) involves confirming the identity of a speaker to ensure they are who they claim to be. In contrast to text-dependent speaker verification, where it is required to say exactly the enrolled or given passphrase, text-independent speaker verification allows users to speak freely without constraints on speech content. This technology, also known as voice biometrics, plays a crucial role in various applications for personnel authentication. Typically, an embedding vector is extracted from enrollment and test speech recordings and processed by a backend classifier, like cosine similarity, to determine whether the recordings are from the same or different speakers.

Since the introduction of the d-vector framework [1], deep learning architectures have been widely used as embedding extractors to capture speaker-specific information. The training of embedding extractors is typically performed using a supervised approach, which relies on a large amount of labeled data. The x-vector paradigm [2], utilizing a Time-Delay Neural Network (TDNN) and statistics pooling, became the leading approach for text-independent speaker verification due to its superior performance. Several variations of the x-vector framework [3]–[5] have also been proposed to reduce speaker verification error rates. Similarly, deep convolutional neural networks (CNNs), especially ResNet-based networks [6], and Long Short-Term Memory (LSTM) network have also been adopted as backbones for SV systems [7]–[10]. Research [11]–[14] has also found that the robustness of x-vectors can be improved by integrating residual connections between the frame-level layers. In [15], the TitaNet speaker embedding learning model was introduced, which employs 1D depth-wise separable convolutions and utilizes the global context of squeeze-and-excitation layers combined with channel attention pooling to extract utterance-level embeddings. Integration of diffusion model with the ECAPA-TDNN paradigm [14] has been investigated in [16].

In efforts to leverage the complementary speaker information captured by different neural architectures [17], [18], several hybrid approaches have emerged, which integrate LSTM or CNN into the x-vector framework and have shown significant improvements in speaker verification performance [19]–[21]. Several hybrid architectures that build upon ECAPA-TDNN have been proposed [22]–[24], incorporating a 2D-CNN before the ECAPA-TDNN to enhance the learning of frame-level features. The ECAPA-CNN-TDNN was proposed by adding a 2D-CNN-based front-end with frequency-wise SE layers to incorporate frequency translational invariance [22]. The MFA-TDNN approach [23] employed Multi-scale Frequency-channel Attention (MFA) preceding ECAPA-TDNN to harness multi-scale processing capability and focus the attention module on the most pertinent features. The SKA-TDNN hybrid framework presented in [24] comprises both frequency-channel-wise SKA-based frontend and multi-scale SKA (msSKA)-based TDNN networks to enhance speaker verification performance.

Unlike supervised SV, self-supervised SV eliminates the need for large-scale labeled data by leveraging unlabeled speech to learn meaningful speaker representations. Instead of relying on manually annotated speaker identities, self-supervised SV methods utilize contrastive learning, clustering-based pseudo-labeling, or predictive modeling techniques to extract speaker-discriminative features. The self-supervised paradigm enables more scalable and cost-effective speaker verification while maintaining competitive performance com-

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pared to supervised methods. In contrastive learning-based self-supervised SV, the embeddings from the same utterance are trained to be close to each other while pushing away embeddings from different utterances [25]–[28]. Since obtaining segments of the same speaker across different utterances is challenging, data augmentation plays a crucial role in enhancing the performance of contrastive learning-based speaker SV frameworks [25], [28]. The second and most widely adopted strategies for self-supervised speaker embedding learning is to create pseudo-labels using clustering algorithms and, like the supervised SV, train the speaker embedding network using the created pseudo-labels in a discriminative fashion [29]–[34]. But their performance is highly dependent on the accuracy of the clustering algorithms [34], [35].

In this work, our contributions include:

- We propose a new hybrid speaker embedding network, that employs a frequency- and channel-wise Selective Kernel Attention (SKA) into the frame-level feature extraction module based on 2D-CNN. The SKA permits each convolutional layer to adaptively choose kernel size based on data. Particularly, the proposed approach integrates a frequency-channel wise SKA-based 2D-CNN feature extraction module with a frame-level network that consists of both a TDNN-LSTM hybrid module and a fully TDNN module, arranged in a cascade structure.
- In order to capture complementarity of different modules, a Multi-Level Attentive Statistics Pooling (MLASP) is employed to encode speaker information within short time-span and utterance-level context. In the MLASP module, The local statistics are also included alongside the features to be attended to provide additional context, enabling the model to efficiently capture speaker discriminative salient features and patterns present in the input data.
- Furthermore, the proposed approach employs an improved Additive Angular Margin (AAM) Softmax loss function that integrates a dynamic and adaptive label noise cleansing method, termed AdaptiveDrop.
- We perform an ablation study using supervised SV experiments to assess the contribution of each module in the proposed framework.
- We evaluate our proposed approach on the widely used VoxCeleb dataset, benchmarking its performance against state-of-the-art (SOTA) methods in both supervised and self-supervised speaker verification tasks.
- We also evaluate the robustness of our proposed speaker recognition model in language-mismatch scenarios.

II. Proposed Framework

This section describes the proposed hybrid speaker embedding extraction approach for robust text-independent speaker verification. In this approach, we introduce a novel hybrid architecture designed to enhance the learning of salient and speaker-discriminative frame-level descriptors while leveraging the complementarity of various network modules. This architecture incorporates a frequency-channel-wise SKA-enhanced 2D-CNN feature extraction module and a Multi-Level ASP mechanism, utilizing context based on local statistics. The proposed framework, shown in Figure 1, consists of four key modules:

A. Feature extraction module based on 2D-CNN and frequency-channel wise SKA mechanism

The idea behind the Selective Kernel Attention (SKA) mechanism is to allow the model to dynamically select which information to focus on in a data driven manner [36]. Therefore, having drawn inspiration from [24], [36], the proposed approach integrates a frequency-channel wise SKA-based 2D-CNN feature extraction module for bagging the temporal-spectral correlations within the input acoustic features. The feature extraction module, as depicted in Figure 1 by the red dotted rectangle, consists of a preceding & a succeeding 2d-convolution layers and two ResBlock layers. The output of the feature extraction module is shown by the solid red arrow line. Each ResBlock, on the other hand, contains a 2d-convolution layer, a frequency-wise SKA (fwSKA) layer, a channel-wise SKA (cwSKA) layer and a Squeeze and Excitation (SE) layer. The whole block (i.e., ResBlock) is covered by a skip connection.

Each of the fwSKA and cwSKA modules undergoes Split, Fuse, and Select operations to generate frequency-wise and channel-wise attended feature maps, respectively. In the Split operation, the input feature map is first splitted into n branches and fed through the n convolution (conv) layers with predefined different kernel sizes. In the Fuse operation, the multi scale features from the conv layers are fused (element-wise). Finally, in the Select operation, soft attention weights (frequency-wise for fwSKA and channel-wise for cwSKA) are learned over the fused features, which are then used to produce the final feature map by applying a weighted summation over the n different branches.

The proposed framework utilizes the feature extraction module to process the input Mel-FilterBank (MFB) features over which SpecAugment [37] is applied on the fly. SpecAugment applies both time and frequency masking by randomly obscuring a subset of frames in the time domain and channels in the frequency domain to make features robust to partial loss of frequency and time information. By passing the augmented MFB features (after applying SpecAugment) through the feature extraction block, the framework captures frame-level representations that encode both temporal dependencies between local frames and spectral relationships across frequency bins.

B. Frame-level network

The frame-level network of the proposed approach consists of a TDNN-LSTM hybrid module and a fully TDNN module arranged in a cascade, as shown in Figure 1, and aims to extract local descriptors with sufficient temporal information for speaker discrimination. The TDNN-LSTM and fully TDNN modules are represented by blue and green dotted rectangles, respectively, with outputs indicated by corresponding colored arrows.

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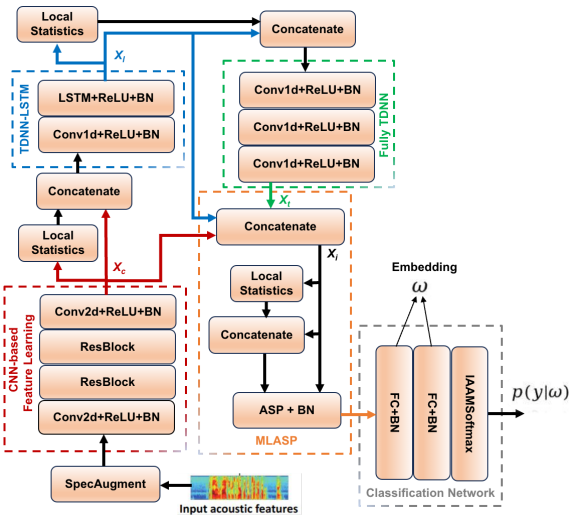


Fig. 1. A schematic diagram showing various steps of the proposed novel hybrid neural speaker embedding network employing frequency- and channel-wise Selective Kernel Attention (SKA) - based feature extraction module and Multi-Level Attention Statistics Pooling (MLASP) mechanism for speaker verification. The 2D-CNN, TDNN-LSTM, fully TDNN, and MLASP modules are depicted by the red, blue, green and orange dotted rectangles, respectively. The Improved Additive Angular Margin Softmax (IAAMSoftmax) objective is used for training the embedding network. Here, FC and BN stands for Fully Connected and Batch Normalization, respectively. We denote the backbone network of this framework (excluding the classification network) as the SKA-HNN (SKA-driven hybrid neural network).

The 2D-CNN-based feature extraction module outputs $\boldsymbol{X}_c \in \mathbb{R}^{d_c \times T}$, where d_c is the dimensionality of 2D-CNN output representations, and T represents the duration in frames. This output, along with local statistics, is concatenated and fed into the TDNN-LSTM hybrid module (comprising one TDNN layer and one LSTM layer) to capture speaker-discriminative information over short and long time spans. During local statistics pooling, 1st and 2nd-order statistics are computed by resampling input sequences and adjusting pooling window shift rates to align with the frame-level features.

The output $\boldsymbol{X}_l \in \mathbb{R}^{d_l \times T}$ from the TDNN-LSTM network, where d_l is its dimensionality, is concatenated with the computed local statistics. This joint representation is passed through the fully TDNN module, consisting of three 1D convolution (TDNN) layers, producing output $\boldsymbol{X}_t \in \mathbb{R}^{d_t \times T}$, where d_t is the dimensionality of the fully TDNN output.

The final outputs, $\boldsymbol{X}_c$, $\boldsymbol{X}_l$, and $\boldsymbol{X}_t$ from the 2D-CNN, TDNN-LSTM, and fully TDNN modules, respectively, are fed to the MLASP (Multi-Level Attention Statistics Pooling) module to aggregate complementary information from these networks. Here, $d_c = 2560$, $d_l = 512$, and $d_t = 512$.

C. Multi-Level Attention Statistics Pooling

At the Multi-Level Attention Statistics Pooling (MLASP) module, the complementarity of diverse network modules is captured by performing a multi-level feature combination (MLFC), where $\boldsymbol{X}_c$, $\boldsymbol{X}_l$, and $\boldsymbol{X}_t$ are concatenated to form $\boldsymbol{X}_i \in \mathbb{R}^{d_i \times T}$, with the integrated feature dimension $d_i = d_c + d_l + d_t = 3584$. The MLFC operation helps to incorporate previously overlooked features, enhancing the robustness of the learned embeddings. A local statistics (LS) pooling operation is applied to $\boldsymbol{X}_i$ to compute the mean $\boldsymbol{\mu}_i$ and standard deviation $\boldsymbol{\sigma}_i$, both matching the dimensionality of $\boldsymbol{X}_i$. These statistics are concatenated with $\boldsymbol{X}_i$, resulting in $\boldsymbol{X}_i^c = [\boldsymbol{X}_i; \boldsymbol{\mu}_i; \boldsymbol{\sigma}_i] \in \mathbb{R}^{3d_i \times T}$. Both $\boldsymbol{X}_i$ and $\boldsymbol{X}_i^c$ are then used by the MLASP module to transform frame-level features into a fixed-dimensional utterance-level vector.

Given frame-level combined representations $\boldsymbol{X}_i = \{\mathbf{x}_i^1, ..., \mathbf{x}_i^T\}$, the formulation of the standard Attention Statistics Pooling (ASP) is defined as follows:

$$\omega = \sum_{t=1}^{T} \alpha_t \mathbf{x}_i^t \tag{1}$$

where t & T represent the frame indice and number of frames, respectively, and $\alpha_t \in [0, 1]$ is a normalized weight, which is computed by

$$\alpha_t = \frac{\exp(e_t)}{\sum_{t=1}^{T} \exp(e_t)}, \quad e_t = \mathbf{v}_t^\intercal \tanh(\mathbf{W}_t \mathbf{x}_i^t + \mathbf{b}_t) \tag{2}$$

where $\mathbf{v}_t$, $\mathbf{W}_t$, and $\mathbf{b}_t$ are trainable parameters and superscript $\intercal$ indicates transpose operation.

By assigning distinct weights for each frame, frame-level representations with relatively higher speaker-relevance can contribute a more significant impact on the embedding vector. In the traditional ASP module, attention weights are learned from the input representations and used to generate attended features via element-wise multiplication with the input.

As depicted in Figure 1, the MLASP module learns attention weights over $\boldsymbol{X}_i^c = [\boldsymbol{X}_i; \boldsymbol{\mu}_i; \boldsymbol{\sigma}_i]$. These weights are then reduced to match the dimensionality of $\boldsymbol{X}_i$ before element-wise multiplication to obtain attended frame-level representations. The attended features are aggregated along the time axis to compute first-order (mean) and second-order (standard deviation) statistics, which are concatenated to form a fixed-dimensional utterance-level vector. In our proposed approach, the rationale for incorporating local statistics alongside the features to be attended is to offer supplementary context, allowing the model to effectively capture speaker-discriminative salient features and patterns inherent in the input data.

D. Classification Network

The classification network, shown in Figure 1, consists of two fully connected (FC) layers and an output softmax layer. The utterance-level representations from the MLASP layer are projected into 192-dimensional speaker embeddings through these FC layers, and the embeddings are then processed by the softmax layer to produce log-probabilities over the set of training speakers. The framework is trained using an improved version of Additive Angular Margin Softmax (AAMSoftmax) loss function. By incorporating a margin m into the cosine similarity computation, the AAMSoftmax enhances intra-class compactness and inter-class separation, leading to more discriminative speaker embeddings [38]. For the detailed formulation of the AAMSoftmax loss function, please refer to

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Appendix A. Our proposed framework utilizes the Improved AAMSoftmax (IAAMSoftmax) loss, which incorporates a dynamic and adaptive label noise cleansing method, called AdaptiveDrop [39], into the widely used AAMSoftmax. Algorithm 1 presents our implementation of IAAMSoftmax (i.e., AAMSoftmax integrated with AdaptiveDrop). Given a training batch X and its corresponding labels (noisy) Y, our approach employs noisy label correction and filtering simultaneously to enhance the label correction process and improve the accuracy of our labels by avoiding wrong corrections. Besides, we alleviate the negative impact of over-cleaning since we first correct suspicious labels before filtering them, which can avoid removing too many samples. To achieve this, we use Cosine Similarity (CS) as a proxy label confidence indicator to detect the mislabeled instances and set a constant CS threshold τ for label filtering to drop high-confident noisy data. First, given each training sample x and its label y, we use the weights of our currently trained embedding network to extract embedding ω of x on the fly at each training step. We then compute cosine similarity $CS(x, y) = \cos(\omega, \omega_y) = \frac{\omega \cdot \omega_y}{\|\omega\| \|\omega_y\|}$ between ω and the current learnt class center/prototype embedding ω_y corresponding to class y. If $CS(x, y) >= \tau$, we retain sample x in the training batch, otherwise it is dropped out from the batch. Indeed, applying filtering after label correction helps improve correction by assuring that the corrected labels also have a high cosine similarity which is a second constraint to ensure higher quality of the corrected labels. Finally, AAMSoftmax loss is computed using the filtered data and labels.

Once training is done, embeddings are extracted from the fully connected layer closest to the pooling layer. Since cosine similarity outperforms PLDA on VoxCeleb test sets when using a margin-based loss function [40]–[42], it is used for verification scoring in the proposed approach. The input/output dimensions of each block in our proposed framework, along with the detailed information flow, are provided in Appendix B and Appendix C.

Algorithm 1 Improved Additive Angular Margin Softmax loss

Inputs: Training data X, labels (noisy) Y, model M, loss function l, cosine similarity threshold τ, epoch to start correcting noisy labels (ESC), epoch to start dropping out noisy labels (ESD).
Output: The best model M.
for epoch $= 1$ **to** max_epochs **do**
  Extract embeddings ω for samples of current batch x.
  Compute cosine similarities (CS) between each sample in x and the class centers ω_i for $i \in \{1, .., C\}$.
  Compute cosine similarities (CS) between each sample in x and its corresponding class center ω_y.
  if $epoch \geq$ ESC **then**
    Correct pseudo-labels y of batch x.
  end if
  if $epoch \geq$ ESD **then**
    Filter samples based on $CS(x, y) = \cos(\omega, \omega_y)$.
    Set $x_{\text{clean}} = \{x_i \text{ for } x_i \in x \text{ if } CS > \tau\}$
    Set $y_{\text{clean}} = \{y_i \text{ for } y_i \in y \text{ if } CS > \tau\}$
  else
    Set $x_{\text{clean}} = x$
    Set $y_{\text{clean}} = y$
  end if
  Compute AAMSoftmax loss l with filtered x_{clean} and y_{clean}.
  Perform gradient descent to update network parameters of model M.
end for

III. Self-Supervised Speaker Verification

Most deep embedding models for speaker verification are trained in a fully supervised manner, relying on large speaker-labeled datasets. However, obtaining a well-annotated dataset is both costly and time-intensive, prompting researchers to explore more efficient self-supervised learning (SSL) approaches. A common strategy to address this challenge is to employ contrastive learning (e.g., SimCLR, MoCo [27], [29], [43]), where embedding networks are pretrained to generate pseudo-labels (PLs), which are then used and refined to train the speaker embedding network in a discriminative manner. Alternatively, clustering algorithms such as K-means, or agglomerative hierarchical clustering can be employed to annotate samples for a subsequent stage of training [32]–[34].

To further evaluate the performance of our proposed hybrid neural approach, we construct a clustering-driven self-supervised speaker verification framework, as illustrated in Figure 2. This setup enables us to simulate label noise and assess the robustness of our architecture on noisy datasets. For this framework, we assume that the VoxCeleb2 development set is unlabeled. The framework presented in Figure 2 comprises three primary modules: the pseudo-label generation module (top), which generates pseudo speaker labels; the embedding network module (middle), responsible for extracting utterance-level embeddings based on these pseudo speaker labels; and the AdaptiveDrop module (bottom) [39], designed to dynamically and adaptively cleanse and mitigate label noise. The AdaptiveDrop module is integrated with AAMSoftmax, and this combined configuration is referred to as Improved AAMSoftmax (IAAMSoftmax). Algorithm 1 details our implementation, including embedding extraction and the filtering of incorrectly annotated data samples.

In the embedding network module, we use the SKA-HNN (selective kernel attention-driven hybrid neural network) backbone architecture, as depicted in Figure 1, as the embedding network. In the pseudo-label generation module, we utilize the CAMSAT algorithm [44] to generate pseudo-labels, using a predefined number of 5,994 clusters. As input features for CAMSAT, we employ 400-dimensional i-vectors [45], [46], which serve as compact, unsupervised speaker representations. These i-vectors help reduce the high dimensionality of MFCC acoustic features, enabling more efficient clustering. In IAAMSoftmax, we integrate AAMSoftmax with the AdaptiveDrop noisy label filtering framework [39] to mitigate noise

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TABLE I
STATISTICS OF THE VOXCELEB 2 DEVELOPMENT SET AND THE VOXCELEB 1 TEST SETS IN TERMS OF NUMBERS OF SPEAKERS, RECORDINGS, TOTAL NUMBER OF TRIALS AND TARGET TRIALS.

Sets	# Speakers	# Recordings	# Trials	# Target trials
VoxCeleb2 (dev)	5994	1092009	NA	NA
VoxCeleb1_O	40	4874	37720	18860
VoxCeleb1_E	1251	145375	581480	290743
VoxCeleb1_H	1190	138137	552536	276270
VoxCeleb1_B	1037	42561	808,574	404287

overfitting during training. To achieve this, we set a cosine similarity (CS) threshold of $\tau = 0.423$ (corresponding to a 65-degree angle) as the maximum allowable deviation between a sample's embedding and its class center, effectively filtering out unreliable and noisy samples. Label noise filtering begins after a 5-epoch warm-up phase.

IV. SUPERVISED SPEAKER VERIFICATION EXPERIMENTS AND RESULTS

A. Experimental setups

The VoxCeleb dataset is used to evaluate the supervised speaker verification performance of the proposed hybrid speaker embedding framework and to compare it with the baseline and state-of-the-art (SOTA) systems. Only the VoxCeleb2 development set is used for training the embedding extractors [47], and results are reported on all three test sets of the VoxCeleb 1 corpus [48]. Additionally, to assess the robustness of speaker recognition models in bilingual scenarios, we conducted evaluations using the VoxCeleb1_B (VoxCeleb1 Bilingual) test set [49], which simulates language-mismatch conditions with numerous cross-lingual trials. Table I provides details on the train-eval/test splits, number of speakers, recordings, and trials. 80-dimensional MFB features are extracted using an analysis window size of 25 ms over a frame shift of 10 ms. The acoustic features are randomly augmented on-the-fly with either MUSAN noise, speed perturbation with rate between 0.95 - 1.05 or reverberation. In addition, SpecAugment is used for applying frequency and time masking on the MFB features.

The embedding networks are trained with segments consisting of 180 frames, using the ADAM optimization technique [50]. The IAAMSoftmax (Improved Additive Angular Margin Softmax) or AAMSoftmax objective was used for training the proposed embedding networks with scale factor $s = 30$ and margin $m = 0.2$. The networks were trained with an initial learning rate 10^{-7} with triangular cyclic scheduling. The batch size for training was set to be 64, which consists of 64 randomly selected speakers. Cosine similarity was used for computing the verification scores in the experiments. Results of our proposed and developed systems are reported in terms of equal error rate (EER) and minimum Detection Cost Function (minDCF) metrics.

B. Experimental results and discussion

1) Ablation Study: In our first set of experiments, we perform ablation study to demonstrate the effectiveness of using the 2D-CNN - based feature learning module, Multi-Level Feature Combination (MLFC), the Local Statistics context-based Multi-Level Attention Statistics Pooling (MLASP) module and IAAMSoftmax loss of the proposed hybrid speaker verification framework. More specifically, we compare the performances of the following systems:

- S0: This is our proposed approach, which is shown in Figure 1 and described in section II.
- S1: Same as our proposed system S0 but utilizes AAM-Softmax instead of Improved AAMSoftmax (IAAMSoftmax).
- S2: Same as system S1 but without utilizing any context in the MLASP module.
- S3: Same as S1 but does not employ MLFC, i.e., for this system $\boldsymbol{X}_i = \boldsymbol{X}_t$. In this case, $\boldsymbol{X}_t$ along with the local statistics computed from it is directly fed to the MLASP module.
- S4: Same as system S1 but does not utilize MLFC (multi-level feature combination) and no context in the MLASP module. For this system, $\boldsymbol{X}_i = \boldsymbol{X}_t$ and the MLASP operation is applied on the top of $\boldsymbol{X}_t$ without appending any local statistics as context.
- S5: Same as system S1 but utilizes vanilla 2D-CNN - based feature extractor instead of the frequency- and channel-wise Selective Kernel Attention (SKA) - based feature extraction module.
- S6: Same as S1 but does not employ any 2D-CNN feature extraction module. In this system, the output of SpecAugment module is directly fed to the TDNN-LSTM module. In this system, the MLASP operation is applied over $\boldsymbol{X}_i = [\boldsymbol{X}_l; \boldsymbol{X}_t]$ after adding local statistics as context.

TABLE II
ABLATION STUDY SHOWING THE EFFECTIVENESS OF USING THE DIFFERENT MODULES OF THE PROPOSED HYBRID SPEAKER VERIFICATION FRAMEWORK. RESULTS ARE REPORTED ON THE VOXCELEB1_O TEST SET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD.

System	Parameters (Million)	Scoring	EER (%)↓	minDCF↓
S0 (ours)	30.91	Cosine	**0.65**	**0.0430**
S1	30.91	Cosine	**0.67**	**0.0442**
S2	29.16	Cosine	0.72	0.0473
S3	26.77	Cosine	0.82	0.0543
S4	26.52	Cosine	0.83	0.0550
S5	17.37	Cosine	0.98	0.0584
S6	6.34	Cosine	1.70	0.1199

The ablation study results, shown in Table II, are presented in terms of the EER and minDCF metrics, and the number of training parameters (in millions). The proposed system (S0) achieved the best performance when incorporating the 2D-CNN-based feature learning, MLFC module, MLASP module with local statistics as context and IAAMSoftmax loss. It outperformed the other six system combinations (S1, S2, S3, S4, S5, and S6) by a significant margin in both EER and minDCF metrics.

Comparing the performance of systems S0 and S1, we observe that system S0 achieved a relative improvement of 3% and 2.7% over S1 in terms of EER and minDCF, re-

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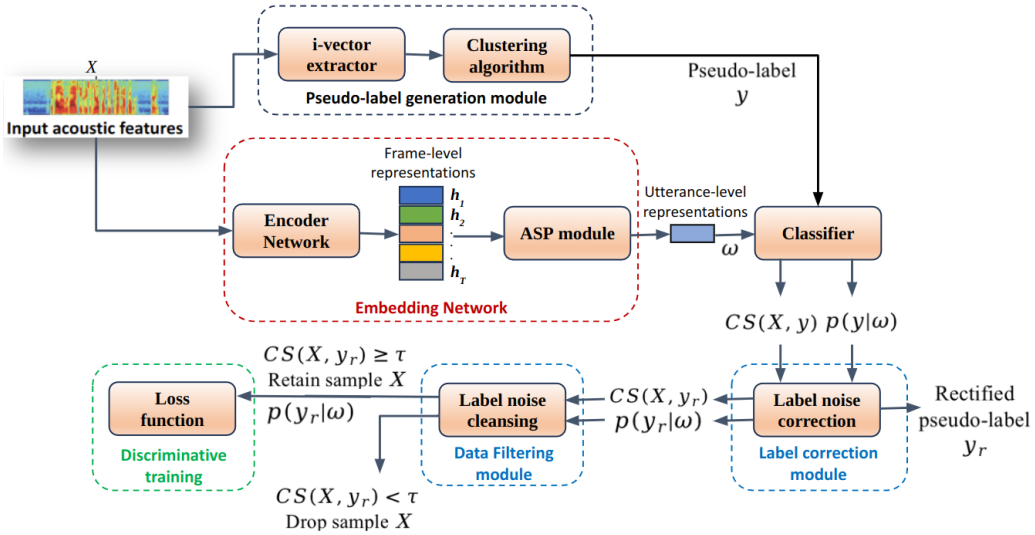


Fig. 2. The General framework for training our self-supervised speaker embedding network using the Improved AAMSoftmax (IAAMSoftmax) loss function, which integrates the AdaptiveDrop method with AAM-Softmax. The pipeline depicts the data flow and the label correction module employed for correcting noisy wrong labels, followed by the data filtering module used to remove the remaining misclassified samples. CS denotes the cosine similarity between the generated sample embedding ω and the class identity center embedding ω_y. τ is the CS threshold. $CS(x, y) = \cos(\omega, \omega_y) = \frac{\omega \cdot \omega_y}{\|\omega\| \|\omega_y\|}$. We use the SKA-HNN backbone network, as depicted in Figure 1, as embedding network.

spectively. This highlights the effectiveness of incorporating IAAMSoftmax in the proposed approach. Comparing systems S0/S1 and S5 shows that the addition of the frequency-channel-wise SKA-based 2D-CNN module significantly improved performance. The value of integrating any 2D-CNN-based feature learning module is further demonstrated by the performance differences between S1, S5, and S6. Besides, in [51], we also demonstrated the superiority of a Selective Kernel Attention (SKA)-driven 2D-CNN frame-level feature learning module over vanilla 2D-CNN, 1D-CNN encoder architectures, and setups without any CNN-based encoder. These findings underscore the effectiveness of attention-augmented CNNs in enhancing frame-level feature extraction, ultimately contributing to more robust and accurate speaker verification systems. System S1 outperformed S2 by approximately 7% in relative EER reduction due to the use of local statistics in the MLASP module. Incorporating the MLFC module in S1 resulted in an 18% relative EER reduction over S3. Conversely, removing the MLFC module and MLASP context from S1, as seen in S4, led to a performance decline. Finally, the ablation study in Table II demonstrated the effectiveness of integrating the frequency- and channel-wise Selective Kernel Attention (SKA)-based feature learning module, the MLFC module, the MLASP module with local statistics context and IAAM-Softmax into the proposed hybrid neural speaker embedding extraction approach. While reducing system complexity lowers the number of training parameters, it often leads to a decline in performance.

2) Performance comparison under language-mismatch conditions: To assess the robustness of our proposed speaker recognition models in bilingual scenarios, we evaluate performance on the VoxCeleb1_B test set. This bilingual test set, introduced in [49], simulates language-mismatch conditions by incorporating a large number of cross-lingual trials. For comparison, we also report results using the ECAPA-TDNN and Disentangled Representation Learning (DRL) [49] frameworks. The DRL approach [49] aims to remove language-dependent information from speaker representations by combining a gradient reversal layer with a mean absolute Pearson correlation minimization objective. As shown in Table III, our proposed approach significantly outperforms the baseline models across both evaluation metrics.

TABLE III
SPEAKER VERIFICATION PERFORMANCES OF THE PROPOSED APPROACH ON THE VOXCELEB1_B TEST SET (I.E., IN LANGUAGE MISMATCH CONDITION) IN TERMS OF EER & MINDCF METRICS. THE BEST RESULTS ARE MARKED IN BOLD.

Test set	Scoring	EER (%)↓	minDCF↓
ECAPA-TDNN	Cosine	4.99	0.324
DRL [49]	Cosine	**3.69**	**0.254**
Proposed (ours)	Cosine	**3.40**	**0.215**

3) Comparison to state-of-the-art: Table IV presents speaker verification results from our proposed hybrid neural speaker embedding model, evaluated using both AAMSoftmax and IAAMSoftmax loss functions across all three test sets of the VoxCeleb1 dataset. The results indicate that the proposed approach achieved state-of-the-art performance across both evaluation metrics. Additionally, the system performed slightly better when using the IAAMSoftmax loss function compared to the AAMSoftmax objective function. However, since the supervised SV task involves minimal label noise in the dataset, the performance gain from IAAMSoftmax over AAMSoftmax remains relatively small (see the results of IAAMSoftmax vs.

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TABLE IV
SUPERVISED SPEAKER VERIFICATION PERFORMANCES OF THE PROPOSED APPROACH USING EITHER IAAMSOFTMAX OR AAMSOFTMAX ON THE THREE TEST SETS OF VOXCELEB1 DATASET IN TERMS OF EER & MINDCF METRICS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD.

Test set	Loss Function	Scoring	EER (%)↓	minDCF↓
VoxCeleb1_O	AAMSoftmax	Cosine	0.67	0.0442
VoxCeleb1_O	IAAMSoftmax	Cosine	**0.65**	**0.0430**
VoxCeleb1_E	AAMSoftmax	Cosine	0.84	0.0544
VoxCeleb1_E	IAAMSoftmax	Cosine	**0.83**	**0.0540**
VoxCeleb1_H	AAMSoftmax	Cosine	1.59	0.0942
VoxCeleb1_H	IAAMSoftmax	Cosine	**1.58**	**0.0939**

AAMSoftmax across all three test sets).

Table V demonstrates the competitiveness of the proposed framework (using IAAMSoftmax loss) by presenting the EER results alongside those of the baseline and SOTA systems on the VoxCeleb1_O, VoxCeleb1_E, and VoxCeleb1_H trial sets. The results indicate that the proposed approach outperforms both the baseline and SOTA systems under all trial conditions.

Impressively, the performance of the proposed framework was able to even consistently surpass the ECAPA-TDNN, ECAPA-CNN-TDNN, MFA-TDNN and SKA-TDNN - based SOTA approaches by a significant margin on all three test sets of the VoxCeleb1 dataset. On VoxCeleb1_O trials condition, our framework attained relative improvements of 35%, 29%, 27% and 23% over the ECAPA-TDNN, ECAPA-CNN-TDNN, MFA-TDNN and SKA-TDNN, respectively, in terms of the EER metric.

On the extended trials (i.e., on VoxCeleb1_E test set) the proposed framework outperformed the ECAPA-TDNN, ECAPA-CNN-TDNN, MFA-TDNN and SKA-TDNN systems by providing a relative improvement of 32%, 25%, 21% and 14%, respectively, in terms of the EER metric. Similarly, on the hard trials setting i.e., on VoxCeleb1_H test set, the proposed approach resulted in better performance in terms of EER metric, than the ECAPA-TDNN, ECAPA-CNN-TDNN, MFA-TDNN and SKA-TDNN systems while ensuring a relative improvement of 31%, 26%, 21% and 15%, respectively. This further substantiates our idea that the complementary attributes inherent in the multi-level attention weights generated from different network modules can allow the learned & extracted embeddings to have more speaker-relevant information. Furthermore, the proposed approach also surpassed SOTA systems on the multi-genre CNCeleb dataset [52], demonstrating its robustness and generalizability across diverse real-world conditions.

V. SELF-SUPERVISED SPEAKER VERIFICATION EXPERIMENTS AND RESULTS

A. Experimental setups

Except for the pseudo-label generation step, which relies on an appropriate clustering algorithm, the self-supervised speaker verification follows a similar experimental setup to the supervised speaker verification, as described in section IV-A.

To evaluate the performance of our proposed approach in a self-supervised speaker verification (SV) setting, we use the CAMSAT clustering algorithm [44], which is based on augmentation mix and self-augmented training, to generate pseudo-labels from the entire VoxCeleb2 development set. We set the number of clusters to 5994. As input features for CAMSAT, we utilize 400-dimensional i-vectors [45], [46], which are compact, unsupervised utterance-level speaker representations. These i-vectors help reduce the high dimensionality of the MFCC (Mel-Frequency Cepstral Coefficients) acoustic features, enabling more efficient clustering. Similar to supervised SV, cosine similarity was used to compute the verification scores in the experiments. The self-supervised SV results for our proposed and developed systems are reported on the VoxCeleb1_O test set in terms of equal error rate (EER) and minimum detection cost function (minDCF) metrics.

TABLE V
SPEAKER VERIFICATION PERFORMANCE ON THE THREE TEST PARTITIONS OF VOXCELEB 1 CORPUS. HERE, AMS STANDS FOR ADDITIVE MARGIN SOFTMAX AND CS STANDS FOR COSINE SIMILARITY.

System	Scoring	EER (%)↓
		Voxceleb1_O
SpecAugment+TDNN [42]	PLDA	2.59
SpecAugment+TDNN+AMS [42]	CS	1.96
SpecAugment+ResNet34 [42]	PLDA	1.68
ECAPA-TDNN	CS	1.00
ECAPA-CNN-TDNN	CS	0.92
DRL [49]	CS	0.99
MFA-TDNN [37], [40]	CS	**0.90**
SKA-TDNN [40]	CS	**0.85**
Proposed (*ours*)	CS	**0.65**
		Voxceleb1_E
SpecAugment+TDNN [42]	PLDA	2.77
SpecAugment+TDNN+AMS [42]	CS	2.31
SpecAugment+ResNet34 [42]	PLDA	1.80
ECAPA-TDNN	CS	1.22
ECAPA-CNN-TDNN	CS	1.11
DRL [49]	CS	1.25
MFA-TDNN [37], [40]	CS	**1.05**
SKA-TDNN [40]	CS	**0.97**
Proposed (*ours*)	CS	**0.83**
		Voxceleb1_H
SpecAugment+TDNN [42]	PLDA	4.83
SpecAugment+TDNN+AMS [42]	CS	4.02
SpecAugment+ResNet34 [42]	PLDA	3.08
ECAPA-TDNN	CS	2.31
ECAPA-CNN-TDNN	CS	2.15
DRL [49]	CS	2.42
MFA-TDNN [37], [40]	CS	**2.00**
SKA-TDNN [40]	CS	**1.87**
Proposed (*ours*)	CS	**1.58**

B. Experimental results and discussion

Table VI presents the self-supervised speaker verification (SV) results obtained using our proposed framework with both AAMSoftmax and Improved AAMSoftmax (IAAMSoftmax) objective functions. In the self-supervised SV setting, our approach with IAAMSoftmax significantly outperformed AAMSoftmax, demonstrating a notable improvement over the supervised SV results.

This performance gain can be attributed to the substantial label noise inherent in the self-supervised SV task, where

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TABLE VI
SELF-SUPERVISED SPEAKER VERIFICATION PERFORMANCES OF THE PROPOSED APPROACH USING THE AAMSOFTMAX AND IAAMSOFTMAX OBJECTIVES ON THE VOXCELEB1_O TEST SET IN TERMS OF EER & MINDCF METRICS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD.

Test set	Loss function	Scoring	EER (%)↓	minDCF↓
VoxCeleb1_O	AAMSoftmax	Cosine	2.122	0.1485
VoxCeleb1_O	IAAMSoftmax	Cosine	**1.920**	**0.1352**

TABLE VII
SOME RECENT SOTA SELF-SUPERVISED SV APPROACHES IN EER (%) COMPARED TO OUR APPROACH ON THE VOXCELEB1_O TEST SET. THE "-" INDICATES THAT RESULTS WERE NOT REPORTED IN THE CORRESPONDING REFERENCE.

SSL Objective	EER (%)↓	minDCF↓
MoBY [27]	8.2	-
InfoNCE [29]	7.36	-
MoCo [53]	7.3	-
ProtoNCE [27]	7.21	-
PCL [27]	7.11	-
CA-DINO [54]	3.585	-
i-mix [35]	3.478	-
l-mix [35]	3.377	-
Iterative clustering [29]	3.09	-
CAMSAT [44]	3.065	0.2155
Subcenter-BoundaryFace [55]	2.609	0.1761
AdaptiveDrop [39]	2.407	0.1668
Proposed (*ours*) (with IAAMSoftmax)	**1.92**	**0.1352**

pseudo-labels generated using our clustering algorithm from the dataset are often noisy. IAAMSoftmax is specifically designed to mitigate the impact of such label noise, leading to a more robust self-supervised speaker representation. As shown in Table VI, the use of IAAMSoftmax resulted in a relative improvement of approximately 9% and 8% in terms of Equal Error Rate (EER) and minimum Detection Cost Function (minDCF), respectively, compared to AAMSoftmax.

Table VII showcases the self-supervised speaker verification results obtained using our proposed hybrid neural speaker embedding model, which employs the IAAMSoftmax loss function. Additionally, the table includes comparisons with other baseline and state-of-the-art (SOTA) self-supervised approaches, all evaluated on the original test set of the VoxCeleb1 dataset.

The presented results in Table VII clearly demonstrate that our proposed approach outperformed all the baselines and delivered state-of-the-art performance across both evaluation metrics with an EER of 1.92% and a minDCF of 0.1352 on the VoxCeleb1_O test set.

VI. CONCLUSION

In this work, we presented a novel hybrid neural speaker embedding network that integrates a frequency-channel-wise selective kernel attention (SKA)-based 2D-CNN encoder module with a frame-level network. The frame-level network consists of a TDNN-LSTM hybrid module followed by a fully TDNN module in a cascaded architecture. To enhance speaker representation, we employed a multi-level attentive statistics pooling (MLASP) mechanism that incorporates local statistics-driven contextual information, enabling the encoding of speaker characteristics at both short-time and utterance levels. Furthermore, our proposed method utilized an improved Additive Angular Margin Softmax (AAMSoftmax) loss function, which incorporates an adaptive and dynamic label noise filtering framework to mitigate the adverse effects of noisy labels and prevent overfitting during training. We carried out extensive supervised and self-supervised speaker verification experiments on the VoxCeleb corpus. The results on the VoxCeleb1 test sets demonstrated that our framework achieves superior speaker verification performance compared to state-of-the-art methods. Additionally, our approach exhibited robust performance under language-mismatch conditions, highlighting its generalization capability across diverse linguistic settings.

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APPENDIX

APPENDIX A: ADDITIVE ANGULAR MARGIN SOFTMAX (AAMSOFTMAX) LOSS

The additive angular margin softmax (AAMSoftmax) objective, a.k.a ArcFace [56], is one of the most popular methods

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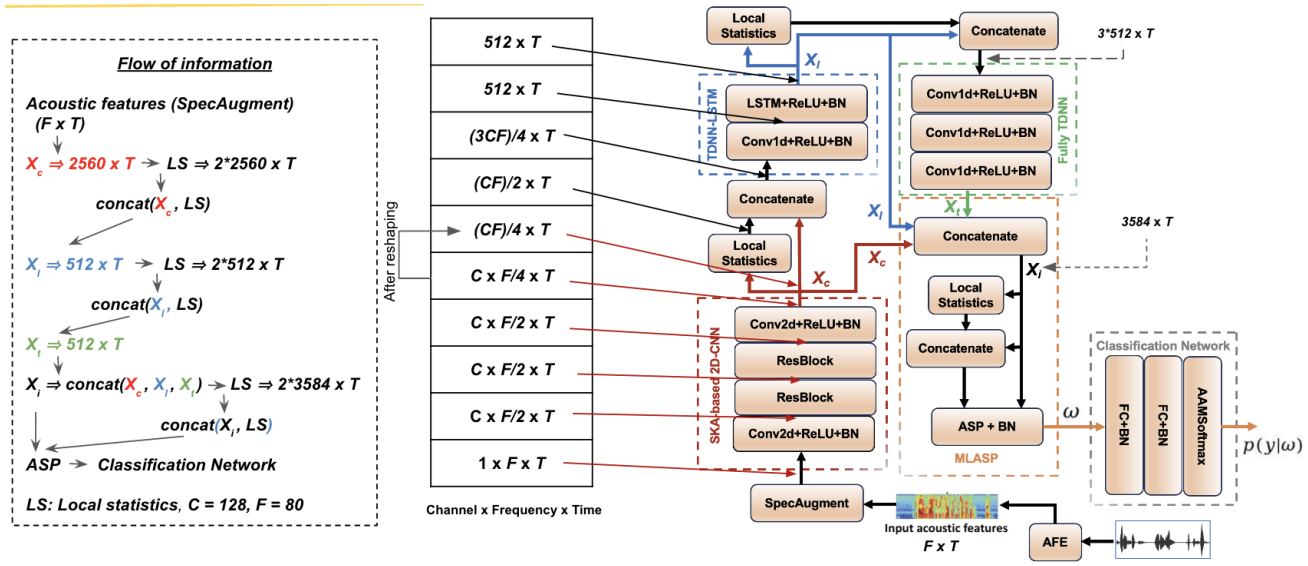


Fig. 3. On the right side, a schematic diagram of the proposed hybrid neural speaker embedding extractor using the AAMSoftmax objective function is presented. The input and output shapes of various modules and layers are shown in the center, while the order of information flow is depicted on the left. Here, F represents the dimensionality of the acoustic features, while T denotes the number of frames. FC and BN stands for Fully Connected and Batch Normalization, respectively. 192-dimensional embeddings are extracted from the FC layer closest to the ASP (attentive statistics pooling) layer.

for training a speaker embedding network. The AAMSoftmax objective is formulated as follows:

$$L_{AAMSoftmax} = -\frac{1}{N}\sum_{i=1}^{N} log(\frac{e^{s(cos(\theta_{y_i}+m))}}{K_1}), \qquad (3)$$

where $K_1 = e^{s(cos(\theta_{y_i}+m))} + \sum_{j=1,j\neq i}^{C} e^{scos\theta_j}$, N is the batch size, C is the number of classes, y_i corresponds to label index, θ_{y_i} represents the angle between the column vector of weight matrix $W_{y_i} \in \mathbb{R}^{192\times 1}$ of the ground truth center and the ith feature embedding $x_i \in \mathbb{R}^{192\times 1}$, where both W_{y_i} and x_i are l_2 normalized. $\theta_j = \arccos(W_j{}^T x_i)$ is the angle between x_i and the jth class center $W_j \in \mathbb{R}^{192\times 1}$. The scale factor s is used to make sure the gradient is not too small during the training and m is a hyperparameter that encourages the similarity of correct classes to be greater than that of incorrect classes by a margin m.

Appendix B: Hybrid Neural Embedding Extractor

Figure 3 depicts a schematic diagram of the Proposed Hybrid Neural Embedding Extractor (using AAMSoftmax loss), the input & output shapes of our different modules/layers, and the order of information flow.

Appendix C: Configuration of the Proposed Hybrid Neural Embedding Extractor (SKA-HNN)

TABLE VIII
THE SKA-HNN ARCHITECTURE. T INDICATES THE DURATION OF FEATURES IN NUMBER OF FRAMES AND $F = 80$ THE FEATURE VECTOR DIMENSIONALITY AND N_c IS THE NUMBER OF CLASSES. ASP, FC & BN STAND FOR ATTENTIVE STATISTICS POOLING, FULLY CONNECTED LAYER AND BATCH NORMALIZATION, RESPECTIVELY. HERE, NUMBER OF CHANNELS IS $C = 128$ AND OUTPUT FEATURES DIMENSION OF HYBRID TDNN-LSTM/FULLY TDNN MODULE $C_1 = 512$.

Layer	*Input Dimension*	*Output Dimension*
Conv2d+ReLU+BN	$F \times$ T	$C \times F/2 \times T$
ResBlock	$C \times F/2 \times T$	$C \times F/2 \times T$
ResBlock	$C \times F/2 \times T$	$C \times F/2 \times T$
Conv2d+ReLU+BN	$C \times F/2 \times T$	$CF/4 \times T \Rightarrow X_c$
Local Statistics	$CF/4 \times T$	$CF/2 \times T$
Concatenation	$CF/2 \times T$ & $CF/4 \times T$	$3CF/4 \times T$
TDNN-LSTM hybrid module	$3CF/4 \times T$	$C_1 \times T \Rightarrow X_l$
Local Statistics	$C_1 \times T$	$2C_1 \times T$
Concatenation	$C_1 \times T$ & $2C_1 \times T$	$3C_1 \times T$
Fully TDNN module	$3C_1 \times T$	$C_1 \times T \Rightarrow X_t$
Concatenation (X_c, X_l, X_t)	$C_1 \times T$ & $C_1 \times T$ & $CF/4 \times T$	$3584 \times T \Rightarrow X_i$
Local Statistics	$3584 \times T$	$2 * 3584 \times T$
Concatenation	$3584 \times T$ & $2 * 3584 \times T$	$3 * 3584 \times T$
ASP+BN (mean & std)	$3 * 3584 \times T$ & $3584 \times T$	$2 * 3584 \times 1$
FC + BN	$2 * 3584 \times 1$	192×1
FC + BN	192×1	192×1
IAAMSoftmax	192×1	$N_c \times 1$

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